

Weekly Internship Report (13/6/26 - 19/6/26)

WEEK 5

Intern's Name: Atharv Gupta

Internship Week: Week 5 (13/6/26 - 19/6/26)

Company/Organization Name: Red Kross Research Foundation

Mentor's Name: Shivam Mittal & Ojas sir

Week Overview:

- Performed security testing on Metasploitable environment focusing on FTP and Telnet services.
- Conducted network reconnaissance, service enumeration, authentication testing, vulnerability assessment, and reporting.
- Configured and tested DVWA (Damn Vulnerable Web Application) for web application security assessment.
- Performed SQL Injection testing, authentication testing, input validation testing, and vulnerability analysis.
- Studied insecure coding practices and analyzed security impacts of identified vulnerabilities.
- Researched and documented 20 recently published CVEs from 2026 covering vulnerabilities in Chrome, Linux Kernel, WordPress, Dell, Cisco, UniFi OS, Yamaha, and other platforms.

Learning Outcomes:

- Gained practical experience in network reconnaissance and service enumeration.
- Learned FTP and Telnet security testing methodologies.
- Improved understanding of vulnerability assessment and penetration testing workflows.
- Developed hands-on experience with SQL Injection testing and web application security analysis.
- Learned how security researchers analyze, classify, and document CVEs.
- Enhanced understanding of real-world vulnerabilities including RCE, SQL Injection, Command Injection, Authentication Bypass, Memory Leaks, and Information Disclosure issues.

Challenges Faced:

- Understanding service enumeration and post-exploitation verification processes.
- Analyzing different vulnerability categories and their security impact.
- Researching and documenting multiple CVEs within a limited timeframe.

Goals for Next Week:

- Study **Rubber Ducky Payloads and USB Attack Vectors**.
- Learn how USB-based attacks are executed and detected.
- Explore **Lock Picking and Physical Security Fundamentals**.
- Understand physical access threats and security controls.

- Analyze the relationship between physical security and cybersecurity.
- Continue improving practical security assessment and research skills.

Mentor Feedback and Suggestions:

- Continue strengthening practical security testing and vulnerability analysis skills.
- Focus on proper reporting and documentation practices.
- Maintain consistency in cybersecurity research and hands-on learning.

Additional Notes:

- This week provided valuable hands-on exposure to network security testing, web application security assessment, vulnerability research, and CVE analysis.

Student Diary (Log) Entry

Week	Task Assigned	Activities Performed	Key Learnings	Additional Remarks
1	Cybersecurity Fundamentals and Research Tasks	Malware analysis, GitHub repository setup, Dridex research, Steganography challenge, Cryptography exercises, AI SOC and ZTNA discussions	Learned malware analysis, hashing vs encryption, cryptographic workflows, GitHub documentation, and cybersecurity research concepts	Improved practical understanding of cybersecurity operations and research methodologies
2	Threat Intelligence, Container Security and Web Security Research	Studied MD5 hash extraction, Threat Intelligence concepts, Docker and Docker Networking, Kubernetes basics, Ransomware workflow, OSINT and CSINT, Threat Intelligence Framework, GHDB research, Burp Suite and FoxyProxy, and performed XXE Injection vulnerability analysis using Burp Suite	Learned threat intelligence concepts including CVE, TTPs, Pyramid of Pain, threat actors, OSINT/CSINT methodologies, container technologies, web vulnerability assessment, and XXE exploitation techniques	Enhanced practical cybersecurity research, threat analysis, reconnaissance, and web application security skills
3	Web Application Security Assessment and Penetration Testing	Network Security Assessment, OWASP ZAP security testing, Prototype Pollution labs, Web LLM Attack labs, and web application penetration testing exercises	Learned prototype pollution, DOM XSS, OWASP ZAP usage, Web LLM attack concepts, and penetration testing methodologies	OWASP ZAP usage, Web LLM attack concepts, and penetration testing methodologies Strengthened practical web application security and vulnerability assessment skills
4	OSINT, Threat Intelligence and Security Research	Researched OSINT tools, lock mechanisms, phishing analysis, OpenDNS phishing quiz, Metasploitable environment, vulnerability report research, and GRC concepts	Learned OSINT reconnaissance, data leakage identification, phishing detection, vulnerable environment analysis, and enterprise security governance	Improved analytical and cybersecurity research capabilities
5	Network Security Testing, Web Application Security	Performed Metasploitable FTP/Telnet testing, DVWA vulnerability assessment, SQL	Learned service enumeration, vulnerability assessment, SQL Injection testing, CVE analysis,	Improved practical penetration testing and cybersecurity research skills

	Testing & CVE Research	Injection testing, and researched 20 recent CVEs	and security reporting methodologies	
--	---------------------------	---	---	--